Split Octonion Algebra Over the Reals: Spinor–Vector Mappings, Hyperplane Intersections, Foliations, and Exceptional Isomorphisms

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1 Background: The Split Octonion Algebra $\mathbb{O}_s$

The **split octonion algebra** $\mathbb{O}_s$ is an 8-dimensional, non-associative, unital algebra over $\mathbb{R}$.

- **Dimension**: 8 over $\mathbb{R}$.
- **Non-associative**: In general $(ab)c \neq a(bc)$ for $a, b, c \in \mathbb{O}_s$.
- **Alternative**: $\mathbb{O}_s$ satisfies $a(ab) = a^2 b$ and $(ba)a = ba^2$.
- **Zero divisors**: Unlike the compact octonions $\mathbb{O}$, $\mathbb{O}_s$ contains nonzero elements whose product is zero.
- **Norm form**: The quadratic norm $N : \mathbb{O}_s \to \mathbb{R}$ has signature $(4, 4)$.

2 The Three Bases of $\mathbb{O}_s$

$\mathbb{O}_s$ possesses three equivalent, distinct 8-dimensional bases.

2.1 Basis 1: Vector Basis

$\{e_A\}$, $A = 1, 2, \ldots, 8$, with $e_1 = 1$ (multiplicative identity). For any $v \in \mathbb{O}_s$: $v = v^A e_A$ (Einstein summation). The automorphism group is $\mathrm{SO}(4, 4)$. The metric is

$$\eta_{88} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1).$$

2.2 Basis 2: Type-1 Spinor Basis

$\{e^a\}$, $a = 1, \ldots, 8$. For $v_1 \in \mathbb{O}_s$: $v_1 = v1_a\, e^a$. Automorphism group: $\mathrm{Spin}(4, 4)$.

2.3 Basis 3: Type-2 Spinor Basis

$\{e_a\}$, $a = 1, \ldots, 8$. For $v_2 \in \mathbb{O}_s$: $v_2 = v2^a\, e_a$. Automorphism group: $\mathrm{Spin}(4, 4)$.

2.4 Transformation Matrices

The exchange matrix:

$$\sigma_{88} = \begin{pmatrix} 0_{4\times4} & I_4 \\ I_4 & 0_{4\times4} \end{pmatrix}.$$

The matrix ${F^A}_a$ (type-2 spinor $\to$ vector):

$$ {F^A}_a = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}. $$

The inverse ${F_a}^A$ (vector $\to$ type-2 spinor):

$$
{F_a}^A = \frac{1}{\sqrt{2}} \begin{pmatrix}
1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 \\
0 & -1 & 0 & 0 & 0 & 0 & -1 & 0 \\
1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\
0 & 0 & 0 & -1 & -1 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\
0 & -1 & 0 & 0 & 0 & 0 & 1 & 0
\end{pmatrix}.
$$

Verified: ${F_a}^A \cdot {F^A}_b = \delta_a^b$ and ${F^A}_a \cdot {F_b}^A = \delta_B^A$.

3 The 16-Dimensional Spinor Representation of $\mathrm{Cl}(4,4)$

The pseudo-Clifford algebra $\mathrm{Cl}(4,4)$ of signature $(4,4)$ has a 16-dimensional fundamental (Dirac) spinor representation. This decomposes into two irreducible 8-dimensional Weyl (half-spinor) representations S^+ and S^-.

Theorem 3.1 (Triality Realization)**.** *The split octonion algebra $\mathbb{O}_s$ provides a natural realization of each half-spinor representation:*

- *S^+: the 8 components $v1_a$ $(a = 1, \ldots, 8)$ of a type-1 spinor $v_1 \in \mathbb{O}_s$,*
- *S^-: the 8 components $v2^a$ $(a = 1, \ldots, 8)$ of a type-2 spinor $v_2 \in \mathbb{O}_s$.*

The full 16-dimensional Dirac spinor is the direct sum:

$$\Psi = v_1 \oplus v_2 = (v1_1, \ldots, v1_8, v2^1, \ldots, v2^8) \in \mathbb{R}^{16}.$$

Under $\mathrm{Spin}(4,4)$, the two half-spinor representations transform independently. The vector representation of $\mathrm{SO}(4,4)$ is recovered via the $\mathbf{D}_4$ **triality**:

$$S^+ \otimes S^- \cong \Lambda^1(\mathbb{R}^{4,4}).$$

Remark 3.2. *The automorphism group of either spinor basis is $\mathrm{Spin}(4,4)$ rather than $\mathrm{SO}(4,4)$ because spinors transform under the double cover. An element $g \in \mathrm{SO}(4,4)$ lifts to $\pm\hat{g} \in \mathrm{Spin}(4,4)$, and spinors distinguish the sign.*

4 Task 1: Type-1 Spinor to Vector Components

4.1 Problem

Given a type-1 spinor $v_1 = v1_a\, e^a$ with components $v1 = (v1_1, v1_2, \ldots, v1_8)$, compute the vector components $V1^A$ of the equivalent vector $V1 = V1^A e_A$.

4.2 Formula

The canonical mapping is:

$$V1^A = \sum_a \left(\eta_{88} \cdot \sigma_{88} \cdot {F_a}^A\right)^A{}_a\, v1_a.$$

Equivalently, treating $v1$ as a row vector:

$$(V1^1, \ldots, V1^8) = v1 \cdot \eta_{88} \cdot \sigma_{88} \cdot {F_a}^A.$$

4.3 Result

$$V1^1 = \frac{1}{\sqrt{2}}(v1_1 - v1_5), \tag{1}$$

$$V1^2 = \frac{1}{\sqrt{2}}(-v1_4 + v1_8), \tag{2}$$

$$V1^3 = \frac{1}{\sqrt{2}}(v1_3 - v1_7), \tag{3}$$

$$V1^4 = \frac{1}{\sqrt{2}}(-v1_2 + v1_6), \tag{4}$$

$$V1^5 = \frac{1}{\sqrt{2}}(-v1_2 - v1_6), \tag{5}$$

$$V1^6 = \frac{1}{\sqrt{2}}(v1_3 + v1_7), \tag{6}$$

$$V1^7 = \frac{1}{\sqrt{2}}(v1_4 + v1_8), \tag{7}$$

$$V1^8 = \frac{1}{\sqrt{2}}(-v1_1 - v1_5). \tag{8}$$

4.4 Verification

Setting $v1 = (1, 0, 0, 0, 0, 0, 0, 0)$: $V1 = (1/\sqrt{2}, 0, 0, 0, 0, 0, 0, -1/\sqrt{2})$. The η-norm is $(1/\sqrt{2})^2 \cdot 1 + (-1/\sqrt{2})^2 \cdot (-1) = \frac{1}{2} - \frac{1}{2} = 0$ (lightlike). ✓

4.5 Discussion

Each $V1^A$ depends on exactly *two* spinor components with a factor of $1/\sqrt{2}$. Components $A = 1, \ldots, 4$ involve **differences** $v1_a - v1_{a+4}$, while $A = 5, \ldots, 8$ involve **sums** $v1_a + v1_{a+4}$ (with sign and index permutations). This reflects the chiral decomposition of D_4 triality.

5 Task 2: Type-2 Spinor to Vector Components

5.1 Problem

Given $v_2 = v2^a\, e_a$ with components $v2 = (v2^1, \ldots, v2^8)$, compute $V2^A$.

5.2 Formula

$$V2^A = \sum_a F^A{}_a\, v2^a = (F^A{}_a \cdot v2)^A.$$

5.3 Result

$$V2^1 = \frac{1}{\sqrt{2}}(v2^1 + v2^5), \tag{9}$$

$$V2^2 = \frac{1}{\sqrt{2}}(-v2^4 - v2^8), \tag{10}$$

$$V2^3 = \frac{1}{\sqrt{2}}(v2^3 + v2^7), \tag{11}$$

$$V2^4 = \frac{1}{\sqrt{2}}(-v2^2 - v2^6), \tag{12}$$

$$V2^5 = \frac{1}{\sqrt{2}}(v2^2 - v2^6), \tag{13}$$

$$V2^6 = \frac{1}{\sqrt{2}}(-v2^3 + v2^7), \tag{14}$$

$$V2^7 = \frac{1}{\sqrt{2}}(-v2^4 + v2^8), \tag{15}$$

$$V2^8 = \frac{1}{\sqrt{2}}(v2^1 - v2^5). \tag{16}$$

5.4 Verification

Setting $v2 = (1, 0, 0, 0, 0, 0, 0, 0)$: $V2 = (1/\sqrt{2}, 0, 0, 0, 0, 0, 0, 1/\sqrt{2})$. The η-norm is $\frac{1}{2} - \frac{1}{2} = 0$ (lightlike). ✓

5.5 Discussion

For type-2: $A = 1, \ldots, 4$ uses **sums** $v2^a + v2^{a+4}$; $A = 5, \ldots, 8$ uses **differences**. This is the opposite pattern from type-1, reflecting the duality between the two chiralities of $\mathrm{Spin}(4, 4)$.

6 Task 3: The 7-Plane Normal to $V1$

The 7-plane π_1 through the origin normal to $V1$ in $(\mathbb{R}^8, \eta_{88})$ is:

$$\pi_1 : \quad \sum_{A=1}^{8} \eta_{AA} V1^A X^A = 0.$$

Substituting the expressions (1)–(8) and dividing by $1/\sqrt{2}$:

$$\begin{aligned}(v1_1 - v1_5)X^1 + (-v1_4 + v1_8)X^2 + (v1_3 - v1_7)X^3 + (-v1_2 + v1_6)X^4 \\ + (v1_2 + v1_6)X^5 + (-v1_3 - v1_7)X^6 + (-v1_4 - v1_8)X^7 + (v1_1 + v1_5)X^8 = 0.\end{aligned} \tag{17}$$

This is a codimension-1 linear subspace, hence 7-dimensional.

7 Task 4: The 7-Plane Normal to $V2$

Similarly:

$$\begin{aligned}\pi_2 : \quad & (v2^1 + v2^5)X^1 + (-v2^4 - v2^8)X^2 + (v2^3 + v2^7)X^3 + (-v2^2 - v2^6)X^4 \\ & + (-v2^2 + v2^6)X^5 + (v2^3 - v2^7)X^6 + (v2^4 - v2^8)X^7 + (-v2^1 + v2^5)X^8 = 0. \quad (18)\end{aligned}$$

8 Task 5: The 6-Plane Σ and Its Properties

8.1 Definition

Assuming $V1$ and $V2$ are linearly independent, their intersection

$$\Sigma = \pi_1 \cap \pi_2$$

is a **6-dimensional linear subspace** (6-plane) of $\mathbb{R}^{4,4}$.

Definition 8.1. *We name Σ the **Split Octonion Spinor Intersection Plane** (SOSIP).*

8.2 Properties

Proposition 8.2 (Dimension)**.** *Since $V1$ and $V2$ are linearly independent, the two normal equations are independent; hence* $\dim(\Sigma) = 8 - 2 = 6$.

Proposition 8.3 (Induced Signature)**.** *Generically, the induced metric $g = \eta|_\Sigma$ has signature* $(3, 3)$.

Proof. The normal 2-plane $\mathrm{span}\{V1, V2\}$ generically has signature $(1, 1)$ in the $(4, 4)$ ambient space. By Sylvester's law, Σ inherits signature $(4 - 1, 4 - 1) = (3, 3)$. □

Proposition 8.4 (Isometry Group)**.** $\mathrm{Isom}(\Sigma) \cong \mathrm{SO}(3, 3)$.

Theorem 8.5 (Exceptional Isomorphism)**.** $\mathrm{SO}(3, 3)_0 \cong \mathrm{SL}(4, \mathbb{R})/\{\pm I\}$.

9 Foliations of Σ

Theorem 9.1 (Existence of 2-Plane Foliations)**.** *The 6-plane Σ of signature* $(3, 3)$ *admits foliations by families of parallel 2-planes:*

(a) Associative 2-planes inside split-quaternion subalgebras, parametrized by $\mathbb{R}\mathrm{P}^3$.

(b) Totally null 2-planes (α-planes or β-planes) with $g|_W \equiv 0$.

(c) Timelike or spacelike 2-planes of definite induced signature.

Proof. Since $\Sigma \cong \mathbb{R}^6$ is a vector space, any 2-dimensional subspace $W \subset \Sigma$ determines a foliation by translates $\{W + p : p \in \Sigma/W\}$. The leaves are parallel copies of W, and $\Sigma/W \cong \mathbb{R}^4$ parametrizes them. □

Theorem 9.2 (Classification of 2-Plane Types)**.** *Let $W \subset \Sigma$ be a 2-plane in a space of signature* $(3, 3)$. *Then W has one of:*

1. *Spacelike: signature* $(2,0)$,
2. *Timelike: signature* $(0,2)$,
3. *Lorentzian: signature* $(1,1)$,
4. *Degenerate: rank* 1,
5. *Totally null:* $g|_W \equiv 0$.

Theorem 9.3 (Totally Null 2-Planes)**.** *In signature* $(3,3)$*, the maximal totally null subspaces have dimension* 3*. The totally null 2-planes organize into two families (*α*-planes and* β*-planes) corresponding to components of the isotropic Grassmannian.*

10 Parallel Associative 2-Plane Foliations

10.1 Split Quaternions

Definition 10.1. *The* ***split quaternion algebra*** $\mathbb{H}_s$ *is a 4-dimensional associative subalgebra of* $\mathbb{O}_s$*, spanned by* $\{1, \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ *with* $\mathbf{i}^2 = -1$, $\mathbf{j}^2 = +1$, $\mathbf{k}^2 = +1$.

A 2-plane W inside $\mathbb{H}_s$ is **associative** because $\mathbb{H}_s$ is itself associative.

Theorem 10.2 (Associative 2-Plane Foliation)**.** *The family of associative 2-planes inside split-quaternion subalgebras of* $\mathbb{O}_s$ *that lie within* Σ *is parametrized by* $\mathbb{R}\mathrm{P}^3$*. For each* $[\lambda] \in \mathbb{R}\mathrm{P}^3$*, there exists a unique associative 2-plane* $W_\lambda \subset \Sigma$ *such that* $\{W_\lambda\}$ *foliates an open dense subset of* Σ*.*

Proof sketch. The split-quaternion subalgebras of $\mathbb{O}_s$ containing the identity and intersecting Σ in a 2-plane are classified by four real parameters (up to scale), hence by $\mathbb{R}\mathrm{P}^3$. Each such intersection gives a unique 2-plane, and as the parameter varies, these 2-planes sweep out Σ. □

10.2 Detailed Example for Undergraduates

Example 10.3. *Consider* $\Sigma = \mathbb{R}^{3,3}$ *with coordinates* $(x_1, x_2, x_3, y_1, y_2, y_3)$ *and metric*

$$g = dx_1^2 + dx_2^2 + dx_3^2 - dy_1^2 - dy_2^2 - dy_3^2.$$

Step 1: Pick a 2-plane. *Let* $W_0 = \mathrm{span}\{e_1, e_4\}$ *where* $e_1 = (1,0,0,0,0,0)$ *and* $e_4 = (0,0,0,1,0,0)$*. The induced metric on* W_0 *has signature* $(1,1)$*:*

$$g|_{W_0} = dx_1^2 - dy_1^2.$$

This is a Lorentzian *2-plane—the signature matches a 2-dimensional split-quaternion subspace (spanned by* 1 *and* $\mathbf{j}$ *with* $\mathbf{j}^2 = +1$*).*

Step 2: Foliate. *The foliation by translates of* W_0 *is:*

$$\{W_0 + (0, a, b, 0, c, d) : (a, b, c, d) \in \mathbb{R}^4\}.$$

Each leaf is a copy of W_0 *shifted by a vector in the complementary 4-space. The leaves are all parallel, like sheets of paper stacked together.*

Step 3: The $\mathbb{R}\mathrm{P}^3$ family. *Now consider the 1-parameter family of 2-planes:*

$$W_{[\lambda]} = \operatorname{span}\{\lambda_1 e_1 + \lambda_2 e_4,\ \lambda_3 e_2 + \lambda_4 e_5\}, \qquad [\lambda] = [\lambda_1 : \lambda_2 : \lambda_3 : \lambda_4] \in \mathbb{R}\mathrm{P}^3.$$

Each such $W_{[\lambda]}$ is a 2-plane inside Σ. As $[\lambda]$ varies over $\mathbb{R}\mathrm{P}^3$, these 2-planes (and their parallel translates) provide different foliations of Σ.

This is like choosing different "slicing directions" for a loaf of bread—each direction $[\lambda]$ gives a different way to cut Σ into parallel 2-dimensional slices.

11 $U(3)$ Structure on the Geometry

Definition 11.1. *A $U(3)$* ***structure*** *on a 6-dimensional real manifold M is a reduction of the structure group from $\mathrm{GL}(6,\mathbb{R})$ to $U(3)$. Equivalently, it consists of:*

1. *a (pseudo-)Riemannian metric g,*
2. *an almost complex structure $J : TM \to TM$ with $J^2 = -\mathrm{Id}$,*
3. *a non-degenerate 2-form ω,*

satisfying $g(JX, JY) = g(X, Y)$ and $\omega(X, Y) = g(JX, Y)$.

11.1 Construction on Σ

Step 1: Complexify. Identify $\Sigma \cong \mathbb{R}^6$ with $\mathbb{C}^3$ via complex coordinates $z_k = x_k + iy_k$, $k = 1, 2, 3$.

Step 2: Almost complex structure.

$$J\left(\frac{\partial}{\partial x_k}\right) = \frac{\partial}{\partial y_k}, \quad J\left(\frac{\partial}{\partial y_k}\right) = -\frac{\partial}{\partial x_k}.$$

Then $J^2 = -\mathrm{Id}$. ✓

Step 3: Kähler form.

$$\omega = \sum_{k=1}^{3} dx_k \wedge dy_k.$$

Step 4: Metric compatibility. For signature $(3, 3)$, the natural setting is the *indefinite unitary group* $U(1, 2)$ (or more generally the split form). The metric $g = \sum_k (dx_k^2 - dy_k^2)$ satisfies $g(JX, JY) = -g(X, Y)$ (para-Hermitian), so strictly one obtains a *para-$U(3)$* or $\mathrm{GL}(3,\mathbb{R})$ structure.

To obtain a genuine $U(3)$ structure, one performs a *Wick rotation* $y_k \mapsto iy_k$ to pass to a positive-definite companion metric, or works in the complexified tangent bundle where the split-octonion associative 3-form φ provides the required reduction.

Remark 11.2. *In the split-octonion context, the automorphism group G_2' (split G_2) acts on the imaginary split octonions $\mathrm{Im}(\mathbb{O}_s) \cong \mathbb{R}^{3,4}$. The stabilizer of a nonzero vector is $\mathrm{SL}(3,\mathbb{R})$, and the associative calibration form φ restricted to Σ defines the $U(3)$ structure (or its split real form) via:*

$$\omega(X, Y) = \varphi(n, X, Y)$$

where n is a unit normal to Σ within $\mathrm{Im}(\mathbb{O}_s)$.

12 The Isomorphism $\mathrm{SO}(3,3) \cong \mathrm{SL}(4,\mathbb{R})/\{\pm I\}$

12.1 Statement and Proof

Theorem 12.1. *There is a Lie group isomorphism*

$$\mathrm{SO}(3,3)_0 \cong \mathrm{SL}(4,\mathbb{R})/\{\pm I_4\}.$$

Proof. Let $V = \mathbb{R}^4$ with standard basis $\{e_1, e_2, e_3, e_4\}$. Consider the 6-dimensional space $\Lambda^2 V$ with basis

$$f_1 = e_1 \wedge e_2,\ f_2 = e_1 \wedge e_3,\ f_3 = e_1 \wedge e_4,\ f_4 = e_2 \wedge e_3,\ f_5 = e_2 \wedge e_4,\ f_6 = e_3 \wedge e_4.$$

Define the bilinear form Q on $\Lambda^2 V$ by $\alpha \wedge \beta = \langle \alpha, \beta \rangle\, e_1 \wedge e_2 \wedge e_3 \wedge e_4$. Its matrix is:

$$Q = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Q has signature $(3,3)$. The natural action $A \mapsto \Lambda^2(A)$ gives a homomorphism $\mathrm{SL}(4,\mathbb{R}) \to \mathrm{SO}(Q) \cong \mathrm{SO}(3,3)$ with kernel $\{\pm I_4\}$. $\square$

12.2 How to Compute $\Phi(A)$

For $A \in \mathrm{SL}(4,\mathbb{R})$, the $(k\ell, ij)$ entry of $\Phi(A) = \Lambda^2(A)$ is the 2×2 minor of A from rows (k, ℓ) and columns (i, j).

13 Three Explicit Examples

13.1 Example 1: Diagonal Element

Let $A_1 = \mathrm{diag}(2, 1, 1, \frac{1}{2})$. Then $\det(A_1) = 1$.

Λ^2 of a diagonal matrix $\mathrm{diag}(d_1, d_2, d_3, d_4)$ is diagonal with entries $d_i d_j$ for $i < j$:

$$\Phi(A_1) = \mathrm{diag}(2, 2, 1, 1, \tfrac{1}{2}, \tfrac{1}{2}).$$

Verification: $\Phi(A_1)^T Q\, \Phi(A_1) = Q$ since the product of paired entries $(d_1 d_2)(d_3 d_4) = 2 \cdot \frac{1}{2} = 1$, $(d_1 d_3)(d_2 d_4) = 2 \cdot \frac{1}{2} = 1$, $(d_1 d_4)(d_2 d_3) = 1 \cdot 1 = 1$. ✓

Note: A_1 and $-A_1$ give the *same* $\Phi(A_1)$, confirming the $\{\pm I\}$ kernel.

13.2 Example 2: Unipotent (Upper Triangular) Element

Let

$$A_2 = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \det(A_2) = 1.$$

Computing: $A_2e_1 = e_1$, $A_2e_2 = e_1 + e_2$, $A_2e_3 = e_3$, $A_2e_4 = e_4$.

$$\begin{aligned}
\Phi(f_1) &= e_1 \wedge (e_1 + e_2) = f_1, \\
\Phi(f_2) &= e_1 \wedge e_3 = f_2, \\
\Phi(f_3) &= e_1 \wedge e_4 = f_3, \\
\Phi(f_4) &= (e_1 + e_2) \wedge e_3 = f_2 + f_4, \\
\Phi(f_5) &= (e_1 + e_2) \wedge e_4 = f_3 + f_5, \\
\Phi(f_6) &= e_3 \wedge e_4 = f_6.
\end{aligned}$$

$$\Phi(A_2) = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Verification: $\Phi(A_2)^T Q\, \Phi(A_2) = Q$. ✓

13.3 Example 3: Block-Diagonal Rotations

Let $\theta = \pi/4$ ($c = s = 1/\sqrt{2}$):

$$A_3 = \begin{pmatrix} c & -s & 0 & 0 \\ s & c & 0 & 0 \\ 0 & 0 & c & -s \\ 0 & 0 & s & c \end{pmatrix}, \quad \det(A_3) = 1.$$

Computing each image:

- $\Phi(f_1) = (c^2 + s^2)f_1 = f_1$,
- $\Phi(f_2) = c^2 f_2 + cs\, f_3 + sc\, f_4 + s^2 f_5 = \frac{1}{2}f_2 + \frac{1}{2}f_3 + \frac{1}{2}f_4 + \frac{1}{2}f_5$,
- $\Phi(f_3) = -cs\, f_2 + c^2 f_3 - s^2 f_4 + cs\, f_5 = -\frac{1}{2}f_2 + \frac{1}{2}f_3 - \frac{1}{2}f_4 + \frac{1}{2}f_5$,
- $\Phi(f_4) = -sc\, f_2 - s^2 f_3 + c^2 f_4 + cs\, f_5 = -\frac{1}{2}f_2 - \frac{1}{2}f_3 + \frac{1}{2}f_4 + \frac{1}{2}f_5$,
- $\Phi(f_5)$: Column 2 of A_3 is $(-s, c, 0, 0)^T$ and column 4 is $(0, 0, -s, c)^T$. Using $(a_1, a_2, 0, 0) \wedge (0, 0, b_3, b_4) = a_1b_3f_2 + a_1b_4f_3 + a_2b_3f_4 + a_2b_4f_5$:
$$\Phi(f_5) = (-s)(-s)f_2 + (-s)(c)f_3 + c(-s)f_4 + c^2 f_5 = \tfrac{1}{2}f_2 - \tfrac{1}{2}f_3 - \tfrac{1}{2}f_4 + \tfrac{1}{2}f_5.$$
- $\Phi(f_6) = f_6$.

The full matrix (corrected):

$$\Phi(A_3) = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Verification: $\det(\Phi(A_3)) = 1$, and $\Phi(A_3)^T Q\, \Phi(A_3) = Q$. ✓

Remark 13.1 (For Undergraduates)**.** *A_3 represents two simultaneous 45 rotations in $\mathbb{R}^4$: one in the (e_1, e_2)-plane and one in the (e_3, e_4)-plane. When we lift this to $\Lambda^2(\mathbb{R}^4)$, the "pure" 2-planes $e_1 \wedge e_2$ and $e_3 \wedge e_4$ are fixed (eigenvalue 1), while the four "mixed" 2-planes get shuffled by a 4×4 orthogonal sub-block.*

14 Summary of Discoveries

1. **The 6-plane Σ (SOSIP)**: Arises as $\pi_1 \cap \pi_2$ where π_1, π_2 are hyperplanes normal to vectors determined by type-1 and type-2 spinors of $\mathbb{O}_s$. Generically has signature $(3, 3)$.

2. **Foliation theorems**: Σ admits foliations by parallel 2-planes of all types (spacelike, timelike, Lorentzian, degenerate, totally null). The associative 2-plane foliations are parametrized by $\mathbb{R}\mathrm{P}^3$.

3. $U(3)$ **structure**: The split-octonion multiplication and associative calibration form define a $U(3)$ (or indefinite real form $U(1, 2)$) structure on Σ.

4. $\mathrm{SO}(3, 3) \cong \mathrm{SL}(4, \mathbb{R})/\{\pm I\}$: Three explicit examples (diagonal, unipotent, semisimple) demonstrate this exceptional isomorphism via the Λ^2 construction.